

Microprocessor System Lab2 Check List

- Date: 2024/10/01
 - Please follow the instructions, and report to TAs after finishing check points.
 - If you have any questions or troubles during the experiment, please feel free to ask TAs or the teacher, discussion with classmates is also encouraged.
 - Please refer to the content from class slides, you can also use the resources from the internet.
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Part I. Question

Please refer to the class content and answer the following questions.

Check Points:

1.1: The attached picture is a picture of a 7-segment display marquee using the mobile phone's continuous shooting function. **Please observe the phenomenon of the blurred image caused by afterimages. Explain what cause this phenomenon (why are there afterimages?).** You can also take pictures with your phone in the second part to see what you get.



Part II. Lab2 Experiment

In this experiment, the 8051 controls the 7-segment display and the dot matrix module to achieve the displaying of specific number string or picture, and make a marquee.

Source codes are already attached in the folder, you can try to write the code by yourself or implement it directly to see the result.

Check Points:

- 2.1:** Connect the wires between 8051 and 7-segment display, execute the number sequence of marquee functions in "main.c" located in the seven_segment folder. You can refer to the lecture slides for the appearance of execution.
- 2.2:** Connect the wires between 8051 and dot matrix module, execute the picture of marquee functions in "main.c" located in the matrix folder. You can refer to the lecture slides for the appearance of execution.

Part III. Practice

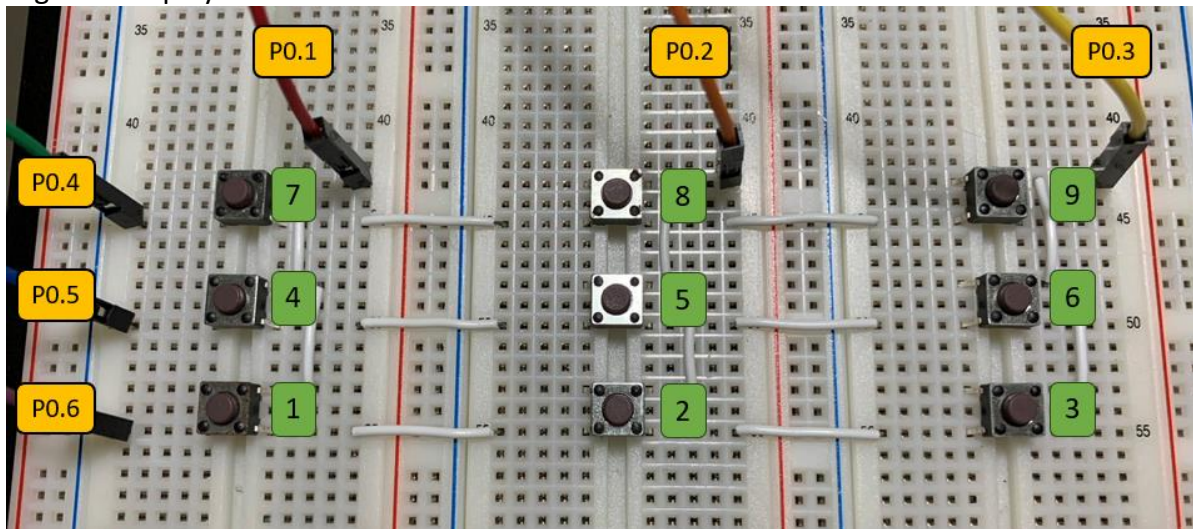
Please refer to the class content and the source code of Part II and complete the Practice. You can finish the practice using the code we've helped you started in the **Practice folder**, or start on your own, as long as it is well functioned.

Description:

This practice contains two parts.

3.1 (seven segment): Implement the behavior of input of a calculator. That is, when the input is 123 then the seven segment should display corresponding "123". This work contains 9

buttons learned in lab1 to generate the corresponding digit (1~9) displayed on the seven segment display.



3.2 (dot matrix): Make a single dot move around the border of a single dot matrix clockwise.

Hint:

3.1: In this part, we give you three TODO to guide you the tasks needed to be done. Extreme cases such as full display isn't required necessarily, any method to deal with it is acceptable.

3.2: In this part, we've finished the display of a single dot part, the rest which includes changing the direction and location of the dot by changing the value in the matrix "m" is needed to be done.

Check Points:

3.1: seven segment display

3.1.1: Functions of the all 9 buttons display are correct

3.1.2: All functions are correct

3.2: dot matrix

3.2.1: All functions are correct